Lab 07 – Risk Matrix Workshop

Industrial Security (Bachelor) • Maps to Section 07 • 60 min, group of 3–4

LAB 07

Goal

Build a defensible cyber risk register for a fictional gas compressor station and use it to argue for three concrete investments.

Step 1 – Define your scales

Define a 5-point **Likelihood** scale and a 5-point **Consequence** scale tailored to a midstream gas operator. Consequence dimensions must include *safety, environment, production loss, regulatory.*

Step 2 – Brainstorm risks

List **ten** cyber risks affecting the station. Examples: ransomware on the Historian, vendor remote-access compromise, USB-borne malware on engineering laptops, SIS bypass via TriStation analogue, etc. Place each in your matrix.

Step 3 – Pick top three

Sort by combined score. For the top three risks, write one Cyber-HAZOP-style row: *guide word, deviation, cyber cause, existing safeguards, gap.*

Step 4 – Investment case

For each top-three risk, propose one mitigation with rough cost (low / medium / high) and time (this quarter / this year / multi-year).

Deliverable

Hand in: the matrix with all ten risks plotted, three Cyber-HAZOP rows, and a 5-line "CFO summary" arguing for the three investments.

Caution

This lab is intentionally fictional. Do not use real plant data, even from public sources – some incidents are still active.